



tive, ensuring the robot can detect small ledges or steps. This feature prevents the robot from unintentionally falling off the edges. We aim to simplify the sensor setup with minimal sensors to facilitate operation in various environments while avoiding complexity, increased development efforts, and higher costs,” elaborates SriKrishna.

The robots have their communication platform developed by SeiAnmai Technologies, which allows the transfer of video and metadata. They use a customised version of Google’s WebRTC, which allows peer-to-peer communication using the server for routing, causing an improvement in latency. This is paired with SeiAnmai’s Tech’s custom UI for controlling the robot.

Currently, only one product designed for research by academia and learners interested in robotics is available as complete robot or as separate parts for self-assembly. This version is entirely open sourced and has an accessible design and software. SriKrishna says he intends to provide access to the software



SriKrishna, Developer and Founder, SeiAnmai Technologies

and designs for individual robot creation. “Currently, the main focus is on developing and open sourcing the software stack. We are considering the GNU open source model, but it is yet to be finalised. We aim for collaborative development and integration of advancements from the user community,” explains SriKrishna.

The startup plans to launch a new robot for commercial use.

It would be targeting hospitals (especially ICU wards), inspections in factories and warehouses, and challenging environments like semiconductor clean rooms and cold storage as possible application areas. SriKrishna reveals his aspirations to combine the existing mobility features with manipulator functionalities such as robotic arms to create a robot capable of moving around and manipulating objects.

The startup is currently supported by I-Hub Foundation for Cobotics (IHFC) IIT Delhi’s Research Entrepreneurship Development for Youth (READY) programme, which provides space, financial support, and resources. However, it acknowledges its openness to collaborations and partnerships for further development.

SriKrishna says that ToTo’s initial deployment is targeted at conferences and events for marketing and data gathering. Future plans involve deploying the robot in more critical scenarios, such as medical centres, after gathering sufficient proof of reliability.

—Yashasvini Razdan

2 PUNE STARTUP GIVES BIRTH TO NEW TWO-WHEELER FORM FACTOR

The logistics sector has become the biggest adopter of electric vehicles in India. However, in the two-wheeler space, the attrition rate due to physical and mental fatigue is alarming for both lease and fleet operators. Can Qargos, with its innovative two-wheeler form factor, help address this issue? Alok Das and Vijay Parveen, the founders of Qargos, explain

Parked at Dassault Systems’ Massachusetts Experience Centre is an electric two-wheeler that looks unlike anything currently on the market! From a distance, it appears larger than a TVS Apache motorcycle, but up close, it is roughly the same size. The key difference? Instead of an engine or batteries, the Qargos F9 houses a

large integrated box within its chassis. Capable of carrying 120kg in this 225-litre space, Pune-headquartered Qargos is targeting the logistics sector with the F9.

Alok Das, co-founder of Qargos, explains, “This is a new form factor in the two-wheeler industry, designed and manufactured in India for global markets.”

The startup aims to deliver 1000 Qargos F9s this year, followed by 20,000 next year, and over 30,000 in the third year. Currently, it is testing the product with major logistics companies in India. For Qargos, the strategy moving forward involves collaborations with lease players and fleet operators.

Vijay Praveen, founder of Qar-